

Lecture 18

The algebraic Bethe ansatz

Objectives. The Bethe equations of [Lecture 15](#), derived a second time without ever writing a wavefunction. One entry of the monodromy matrix creates magnons; the RTT relation says how the diagonal entries move past it; acting with the transfer matrix on a product of creation operators gives the wanted term and a set of unwanted ones, and the unwanted ones cancel exactly when the Bethe equations hold. What is left is the eigenvalue of the whole commuting family, and it satisfies Baxter's T-Q relation: [Lecture 19](#) takes the Bethe equations and those eigenvalues to the thermodynamic limit, [Lecture 20](#) generalises T-Q itself. The same steps, with sinh for linear functions, solve the XXZ chain.

18.1 The monodromy matrix and its reference state

[Lecture 17](#) built, for the Heisenberg chain of N sites, the Lax operator $L_{an}(\lambda) = \lambda + iP_{an}$ (17.3), the monodromy matrix and the transfer matrix

$$T_a(\lambda) = L_{aN}(\lambda) \cdots L_{a1}(\lambda) = \begin{pmatrix} A(\lambda) & B(\lambda) \\ C(\lambda) & D(\lambda) \end{pmatrix}_a, \quad t(\lambda) = A(\lambda) + D(\lambda), \quad (18.1)$$

and proved the RTT relation (17.7), $R_{ab}(\lambda - \mu)T_b(\mu)T_a(\lambda) = T_a(\lambda)T_b(\mu)R_{ab}(\lambda - \mu)$ with $R(\lambda) = \lambda - iP$, and $[t(\lambda), t(\mu)] = 0$ (Theorem 17.4). The Hamiltonian is $H = \frac{J}{2} (i \frac{d}{d\lambda} \ln t|_0 - N)$ and the shift is $t(0) = i^N U$ (Proposition 17.7, Proposition 17.5). The goal is now the spectrum of $t(\lambda)$: one family of eigenvectors, good for every λ at once, and with it the spectrum of H .

The entries A, B, C, D are operators on the chain, and [Figure 18.1](#) says what each does. Read the auxiliary line as carrying a spin from left to right: it enters in the state of the column index and leaves in the state of the row index. At every vertex $L_{an} = \lambda + iP_{an}$ either lets both spins pass or swaps them, so the number of down spins on the auxiliary line and the chain together is conserved: P commutes with S^z of the pair it acts on. Hence A and D conserve S^z of the chain, B (in $\downarrow$, out $\uparrow$) leaves one more down spin on the chain than it found, and C takes one away:

$$[S^z, B(\lambda)] = -B(\lambda), \quad [S^z, C(\lambda)] = C(\lambda), \quad [S^z, A(\lambda)] = [S^z, D(\lambda)] = 0. \quad (18.2)$$

The obvious state to start from is the ferromagnetic one, all spins up, $|0\rangle = |\uparrow\uparrow \cdots \uparrow\rangle$, the $|\uparrow\uparrow\rangle$ of [Lecture 15](#). On it every Lax operator is triangular.

Proposition 18.1 (The reference state). For all λ ,

$$A(\lambda) |0\rangle = (\lambda + i)^N |0\rangle, \quad D(\lambda) |0\rangle = \lambda^N |0\rangle, \quad C(\lambda) |0\rangle = 0, \quad (18.3)$$

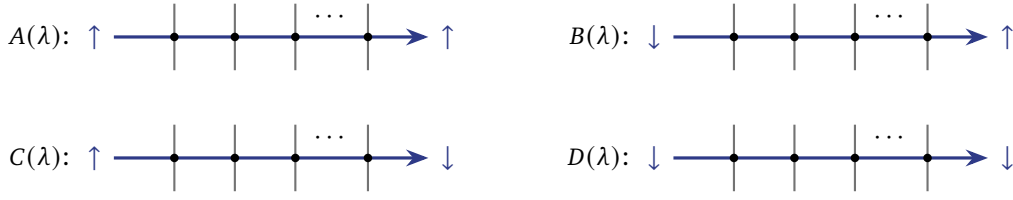


Figure 18.1: The four entries of the monodromy matrix: the row of Figure 17.2 with its two ends fixed, the column index entering on the left (site 1 first) and the row index leaving on the right.

and, with $|n\rangle$ the state with one down spin at site n ,

$$B(v) |0\rangle = i \sum_{n=1}^N v^{n-1} (v+i)^{N-n} |n\rangle = i \frac{(v+i)^N}{v} \sum_{n=1}^N \left(\frac{v}{v+i} \right)^n |n\rangle \quad (v \neq 0, -i). \quad (18.4)$$

Proof. From (17.3), acting on site n in the state $\uparrow$,

$$L_{an}(\lambda) |\uparrow\rangle_n = \begin{pmatrix} (\lambda+i) |\uparrow\rangle_n & i |\downarrow\rangle_n \\ 0 & \lambda |\uparrow\rangle_n \end{pmatrix}_a,$$

since $S^+ |\uparrow\rangle = 0$, $S^- |\uparrow\rangle = |\downarrow\rangle$ and $S^z |\uparrow\rangle = \frac{1}{2} |\uparrow\rangle$. Sites $1, \dots, N$ are acted on by different factors of (18.1), so on $|0\rangle$ the monodromy is a product of triangular 2×2 matrices. Its lower-left entry vanishes and its diagonal entries are the products of the diagonals. The upper-right entry is a sum over the one place where the product steps from the second row to the first: on sites $1, \dots, n-1$ the factor is λ , at site n it is $i |\downarrow\rangle_n$, and on sites $n+1, \dots, N$ it is $\lambda+i$. Set $\lambda = v$. ■

So B creates. For $v \neq 0, -i$, $B(v) |0\rangle$ is a plane wave: its amplitude on site n is proportional to e^{ikn} with

$$e^{ik} = \frac{v}{v+i}, \quad \text{i.e., in the rapidity of (15.5),} \quad v = -\lambda - \frac{i}{2}. \quad (18.5)$$

It is the magnon $|k\rangle$ of (15.4) for every such v , not only for the N values allowed on the ring; periodicity has not been imposed, and $B(v) |0\rangle$ is an eigenvector of H only when $e^{ikN} = 1$. At the two excluded values the vector is not a plane wave at all but a single site, $B(0) |0\rangle = i^N |1\rangle$ and $B(-i) |0\rangle = i(-i)^{N-1} |N\rangle$; they are the ends $k \rightarrow \pm i\infty$ of (18.5), and they return in § 18.3 as the singular pair. The grid point $k = 0$ is not among the finite v at all: it is Lecture 15's root at infinity. The arguments v of the creation operators play the part of rapidities, shifted and reflected. **The reflection is not a choice.** With $v = \lambda - \frac{i}{2}$ instead, $v/(v+i)$ would be e^{-ik} and $B(v) |0\rangle$ the magnon of momentum $-k$; the check script finds the two identifications give orthogonal states whenever the total momentum is not 0 or π .

Which of B and C creates was found by computation, as it must be: in a book with the other ordering of RTT (Remark 17.2) the answer, and every sign below, can differ.

18.2 The exchange relations

To act with $t = A + D$ on a product of B 's, one needs to move A and D to the right, past each B , until they reach $|0\rangle$. The RTT relation is sixteen such rules, one for each entry of a 4×4 matrix in the auxiliary spaces $a \otimes b$.

Lemma 18.2 (*RTT, entry by entry*). Write $T_{11} = A$, $T_{12} = B$, $T_{21} = C$, $T_{22} = D$. For all $i, j, k, l \in \{1, 2\}$,

$$(\lambda - \mu) T_{jl}(\mu) T_{ik}(\lambda) - i T_{il}(\mu) T_{jk}(\lambda) = (\lambda - \mu) T_{ik}(\lambda) T_{jl}(\mu) - i T_{il}(\lambda) T_{jk}(\mu). \quad (18.6)$$

Proof. In the leg notation of Definition 8.1, the $(ij), (kl)$ entry of $T_a(\lambda)T_b(\mu)$ is $T_{ik}(\lambda)T_{jl}(\mu)$, and of $T_b(\mu)T_a(\lambda)$ it is $T_{jl}(\mu)T_{ik}(\lambda)$: the same operators in the other order, which now matters. Multiplying on the right by P_{ab} exchanges the column indices k and l , and conjugating by P_{ab} exchanges the legs, so $P T_b(\mu)T_a(\lambda) = T_a(\mu)T_b(\lambda) P$. With $R = (\lambda - \mu) - iP$, (17.7) reads

$$(\lambda - \mu) T_b(\mu)T_a(\lambda) - i T_a(\mu)T_b(\lambda)P = (\lambda - \mu) T_a(\lambda)T_b(\mu) - i T_a(\lambda)T_b(\mu)P,$$

whose $(ij), (kl)$ entry is (18.6). ■

Three of the sixteen are needed. Write

$$f(u) = \frac{u - i}{u}, \quad g(u) = \frac{i}{u}. \quad (18.7)$$

Proposition 18.3 (*The exchange relations*). For all λ, μ , and for $\lambda \neq \mu$ in the last two,

$$B(\lambda)B(\mu) = B(\mu)B(\lambda), \quad (18.8)$$

$$A(\lambda)B(\mu) = f(\lambda - \mu) B(\mu)A(\lambda) + g(\lambda - \mu) B(\lambda)A(\mu), \quad (18.9)$$

$$D(\lambda)B(\mu) = f(\mu - \lambda) B(\mu)D(\lambda) - g(\lambda - \mu) B(\lambda)D(\mu). \quad (18.10)$$

Proof. Each is one entry of (18.6). With $(i, j, k, l) = (1, 1, 2, 2)$ both products on each side are B 's: $(\lambda - \mu - i)B(\mu)B(\lambda) = (\lambda - \mu - i)B(\lambda)B(\mu)$, which gives (18.8) for $\lambda - \mu \neq i$, and for all λ, μ because both sides are polynomials. With $(1, 1, 1, 2)$, $(\lambda - \mu - i)B(\mu)A(\lambda) = (\lambda - \mu)A(\lambda)B(\mu) - iB(\lambda)A(\mu)$; divide by $\lambda - \mu$. With $(1, 2, 2, 2)$, $(\lambda - \mu)D(\mu)B(\lambda) - iB(\mu)D(\lambda) = (\lambda - \mu - i)B(\lambda)D(\mu)$; exchange the names $\lambda \leftrightarrow \mu$ and divide by $\lambda - \mu$. ■

Read (18.9) as a rule for moving $A(\lambda)$ one step to the right. Either A keeps its argument, and pays $f(\lambda - \mu)$, or A and B exchange arguments, at the price $g(\lambda - \mu)$: the two terms of $R = (\lambda - \mu) - iP$, straight through or swapped. The signs of i depend on the ordering of RTT ; with $i \rightarrow -i$ the last two are false in this course's conventions (the check script). (18.8) has no sign to get wrong, which is why the order of the creation operators is safe.

18.3 Bethe vectors: wanted and unwanted terms

For complex numbers $v_1, \dots, v_M$ define the *Bethe vector*

$$\Psi(v_1, \dots, v_M) = B(v_1)B(v_2) \cdots B(v_M) |0\rangle. \quad (18.11)$$

By (18.8) it is symmetric in the v_j , and by (18.2) it has $S^z = \frac{N}{2} - M$: M down spins, each brought in by one auxiliary line (Figure 18.2). Beyond one magnon (18.4), no wavefunction will be written.

Theorem 18.4 (*Wanted and unwanted terms*). Let $v_1, \dots, v_M$ be pairwise distinct and

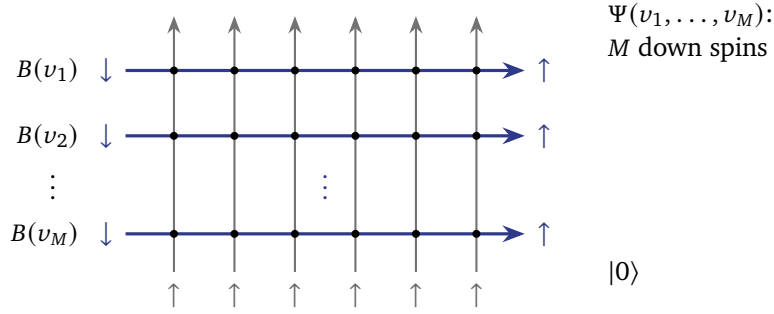


Figure 18.2: A Bethe vector. The reference state (bottom) is all up; each creation operator is a row that enters with a down spin and leaves with an up spin, so it deposits one down spin on the chain, spread over the sites with amplitudes set by its argument. Operators act from the bottom up: $B(v_M)$ first. By (18.8) the order of the rows does not matter.

$\lambda \notin \{v_1, \dots, v_M\}$. Then

$$t(\lambda) \Psi(v_1, \dots, v_M) = \Lambda(\lambda) \Psi(v_1, \dots, v_M) + \sum_{j=1}^M \mathcal{M}_j(\lambda) B(\lambda) \prod_{k \neq j} B(v_k) |0\rangle, \quad (18.12)$$

with

$$\Lambda(\lambda) = (\lambda + i)^N \prod_{j=1}^M \frac{\lambda - v_j - i}{\lambda - v_j} + \lambda^N \prod_{j=1}^M \frac{\lambda - v_j + i}{\lambda - v_j}, \quad (18.13)$$

$$\mathcal{M}_j(\lambda) = \frac{i}{\lambda - v_j} \left[(v_j + i)^N \prod_{k \neq j} \frac{v_j - v_k - i}{v_j - v_k} - v_j^N \prod_{k \neq j} \frac{v_j - v_k + i}{v_j - v_k} \right]. \quad (18.14)$$

The first term of (18.12) is *wanted*: it is a multiple of Ψ . The others are *unwanted*: in each, one creation operator has had its argument replaced by λ (Figure 18.3).

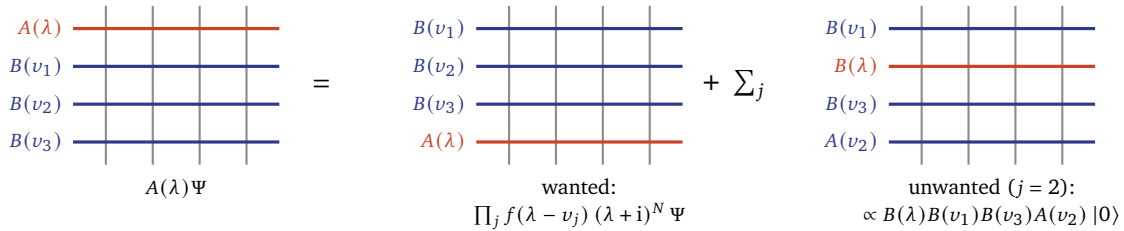


Figure 18.3: Moving $A(\lambda)$ down through the creation rows (here $M = 3$; rows act from the bottom up, as in Figure 18.2). If A keeps its argument at every exchange it reaches $|0\rangle$ as $A(\lambda)$ and returns $(\lambda + i)^N$: the wanted term. If it exchanges arguments with $B(v_j)$ it reaches $|0\rangle$ as $A(v_j)$, returning $(v_j + i)^N$, and the j -th creation row now carries λ : an unwanted term. The argument λ (red) is never lost, only handed on. $D(\lambda)$ does the same with λ^N and v_j^N .

Where do the two kinds of term come from? Each use of (18.9) replaces $A(x)B(y)$ by terms $B(y)A(x)$ and $B(x)A(y)$: the set of arguments $\{x, y\}$ is preserved. Moving $A(\lambda)$ to the right through $B(v_1) \cdots B(v_M)$, every term therefore ends as M creation operators and one A , whose arguments together are $\lambda, v_1, \dots, v_M$. Either A ends with λ , and the B 's make Ψ ; or

it ends with some v_j , and the B 's make $B(\lambda) \prod_{k \neq j} B(v_k)$. The coefficients are easy to guess. The wanted one comes from keeping the argument at every step: $\prod_k f(\lambda - v_k)$. For the j -th unwanted one, use (18.8) to put $B(v_j)$ first: A must exchange at the first step, paying $g(\lambda - v_j)$ and becoming $A(v_j)$, and then keep v_j at every later step, paying $f(v_j - v_k)$.

Lemma 18.5 (Moving A and D through the creation operators). *For pairwise distinct $v_1, \dots, v_M$ and $\lambda \notin \{v_k\}$,*

$$\begin{aligned} A(\lambda) \prod_{k=1}^M B(v_k) &= \prod_k f(\lambda - v_k) \prod_k B(v_k) A(\lambda) \\ &\quad + \sum_{j=1}^M g(\lambda - v_j) \prod_{k \neq j} f(v_j - v_k) B(\lambda) \prod_{k \neq j} B(v_k) A(v_j), \end{aligned} \quad (18.15)$$

and the same for D with $f(u) \rightarrow f(-u)$ and $g(u) \rightarrow -g(u)$ throughout.

Proof (★). The guess put $B(v_j)$ first, a different order for each j , so it does not yet show that one expansion has all these coefficients at once. Induct on M ; $M = 1$ is (18.9). For $M \geq 2$ apply (18.9) to $A(\lambda)B(v_1)$, then the hypothesis for $M - 1$ to both $A(\lambda)$ and $A(v_1)$ acting on $B(v_2) \cdots B(v_M)$. The wanted term and the term ending in $A(v_1)$ come out as claimed. A term ending in $A(v_j)$, $j \geq 2$, arrives twice, with total coefficient $\prod_{k \neq 1, j} f(v_j - v_k)$ times $f(a)g(b) + g(a)g(b - a)$, where $a = \lambda - v_1$, $b = \lambda - v_j$; a line of algebra gives $f(a)g(b) + g(a)g(b - a) = g(b)f(a - b)$, and $f(a - b) = f(v_j - v_1)$ is the missing factor. For D the same identity holds for $f(-u)$ and $-g(u)$, being the one above at $(-a, -b)$. (The check script confirms (18.15) as an operator identity for two and three sites.) ■

Proof of Theorem 18.4. Apply Lemma 18.5 to $|0\rangle$ and use (18.3): $A(\lambda)|0\rangle = (\lambda + i)^N|0\rangle$, $A(v_j)|0\rangle = (v_j + i)^N|0\rangle$, $D(\lambda)|0\rangle = \lambda^N|0\rangle$, $D(v_j)|0\rangle = v_j^N|0\rangle$. Since $f(\lambda - v) = (\lambda - v - i)/(\lambda - v)$ and $f(v - \lambda) = (\lambda - v + i)/(\lambda - v)$, the wanted terms of A and D add to (18.13), and the unwanted coefficients to (18.14). ■

Corollary 18.6 (The algebraic Bethe ansatz). *Call the roots regular if they are pairwise distinct and none is 0 or $-i$. For pairwise distinct $v_1, \dots, v_M$, all the unwanted coefficients vanish if and only if*

$$(v_j + i)^N \prod_{k \neq j} (v_j - v_k - i) = v_j^N \prod_{k \neq j} (v_j - v_k + i), \quad j = 1, \dots, M. \quad (18.16)$$

If they hold, $\Psi(v_1, \dots, v_M)$ is either zero or a simultaneous eigenvector of every $t(\lambda)$, with eigenvalue $\Lambda(\lambda)$, which is then a polynomial. Under $v_j = -\lambda_j - \frac{i}{2}$ (18.5), and when no factor in (15.15) vanishes or diverges, (18.16) is the system (15.15) of Lecture 15.

Proof. M_j is $i/(\lambda - v_j)$ times the difference of the two sides of (18.16), divided by $\prod_{k \neq j} (v_j - v_k) \neq 0$. If (18.16) holds, (18.12) says $t(\lambda)\Psi = \Lambda(\lambda)\Psi$ for $\lambda \notin \{v_j\}$. If $\Psi \neq 0$, pick a component $\Psi_x \neq 0$: then $\Lambda(\lambda) = (t(\lambda)\Psi)_x / \Psi_x$ is a polynomial in λ , and the eigenvalue equation extends to every λ . For the last statement substitute: $v_j + i = -(\lambda_j - \frac{i}{2})$, $v_j = -(\lambda_j + \frac{i}{2})$ and $v_j - v_k \pm i = -(\lambda_j - \lambda_k \mp i)$, so (18.16) becomes $(\lambda_j - \frac{i}{2})^N \prod (\lambda_j - \lambda_k + i) = (\lambda_j + \frac{i}{2})^N \prod (\lambda_j - \lambda_k - i)$, a common sign $(-1)^{N+M-1}$ having cancelled; divide. ■

The Bethe equations of Lecture 15 have come back without a wavefunction, and so have its vectors: for $M = 1$ by (18.4), and in general the check script finds Ψ proportional to (15.14) for every solution it tried ($N = 6, 8$, $M \leq 4$) — not proved here, and not needed.

What the corollary does not say. It does not say $\Psi \neq 0$. The singular pair of § 15.6, $\lambda_{1,2} = \pm \frac{i}{2}$, is $\{v_1, v_2\} = \{0, -i\}$; it satisfies (18.16) (both sides vanish), and $B(0)B(-i)|0\rangle = 0$ (the check script, $N = 4, 6, 8$), although neither factor alone kills $|0\rangle$. Lecture 15's singular eigenstate comes from the regularisation in that lecture's tutorial, not from the corollary — which is why what follows is stated for regular roots. Nor does it give the converse, that an eigenvector forces (18.16): that would need the unwanted vectors to be linearly independent, which is not proved here (for roots that solve nothing the check script finds no eigenvector). Nor does it decide completeness, which remains Lecture 15's cited question.

18.4 The eigenvalue

$\Lambda(\lambda)$ is the eigenvalue of the whole commuting family at once, so the eigenvalues of H and of the shift follow from Proposition 17.7 and Proposition 17.5 without further work.

Proposition 18.7 (Momentum and energy from Λ). *Let $v_1, \dots, v_M$ be regular roots solving (18.16), with $\Psi \neq 0$, $N \geq 2$, and $\lambda_j = -v_j - \frac{i}{2}$ with momenta $e^{ik_j} = (\lambda_j + \frac{i}{2})/(\lambda_j - \frac{i}{2})$. Then*

$$U\Psi = e^{-iK}\Psi, \quad K = \sum_j k_j, \quad \text{and} \quad H\Psi = E\Psi, \quad E = -\sum_{j=1}^M \frac{2J}{4\lambda_j^2 + 1},$$

which is (15.16).

Proof. At $\lambda = 0$ the second term of (18.13) vanishes with its derivative (it carries λ^N , $N \geq 2$, and no pole at 0 since $v_j \neq 0$). So $\Lambda(0) = i^N \prod_j (v_j + i)/v_j$, not zero since $v_j \neq -i$, and

$$\frac{d}{d\lambda} \ln \Lambda|_0 = \frac{N}{i} + \sum_j \left(\frac{1}{v_j} - \frac{1}{v_j + i} \right) = -iN + \sum_j \frac{i}{v_j(v_j + i)}.$$

With $v_j = -\lambda_j - \frac{i}{2}$, $(v_j + i)/v_j = (\lambda_j - \frac{i}{2})/(\lambda_j + \frac{i}{2}) = e^{-ik_j}$ and $v_j(v_j + i) = \lambda_j^2 + \frac{1}{4}$. Now $t(0) = i^N U$ gives the first claim, and (17.11) gives $E = \frac{J}{2} (i \frac{d}{d\lambda} \ln \Lambda|_0 - N) = -\frac{J}{2} \sum_j \frac{1}{\lambda_j^2 + 1/4}$. ■

Everything Lecture 15 extracted from the wavefunction — the Bethe equations, the energy, the momentum — has been obtained from the monodromy matrix and one reference state. The price was paid in Lecture 17: the RTT relation. The return is more than a second derivation. Λ fixes the eigenvalue of every conserved charge at once — that of $\frac{d^k}{d\lambda^k} \ln t|_0$ is $\frac{d^k}{d\lambda^k} \ln \Lambda|_0$ — and the method runs for any model with this R -matrix, or the trigonometric one of § 18.6, and a reference state on which the monodromy is triangular. That includes chains whose sites carry different spectral parameters (reading at the end); for a larger auxiliary space there is more than one kind of creation operator and the ansatz has to be nested — both are what Lecture 22 needs.

18.5 Baxter's T–Q relation

Collect the roots into one polynomial,

$$Q(\lambda) = \prod_{j=1}^M (\lambda - v_j). \quad (18.17)$$

Then $\prod_j (\lambda - v_j \mp i)/(\lambda - v_j) = Q(\lambda \mp i)/Q(\lambda)$, and (18.13) is, multiplied out,

$$\Lambda(\lambda) Q(\lambda) = (\lambda + i)^N Q(\lambda - i) + \lambda^N Q(\lambda + i), \quad (18.18)$$

Baxter's T-Q relation. Baxter found it in 1972 for the eight-vertex model, where the reference-state argument of this lecture does not apply as it stands, by constructing Q as an operator commuting with the transfer matrix. It says the same as (18.13), but it turns the Bethe equations into a statement about polynomials.

Proposition 18.8 (The Bethe equations as analyticity). *Let $v_1, \dots, v_M$ be pairwise distinct and define Λ by (18.18). Then Λ is a polynomial if and only if the v_j satisfy (18.16).*

Proof. Λ is the right side of (18.18) divided by Q , whose zeros are simple. It is a polynomial iff the right side vanishes at every v_j : $(v_j + i)^N Q(v_j - i) + v_j^N Q(v_j + i) = 0$. Since $Q(v_j \mp i) = \mp i \prod_{k \neq j} (v_j - v_k \mp i)$, this is (18.16). ■

The eigenvalue of $t(\lambda)$ on a common eigenvector is a polynomial in λ — the argument in the proof of Corollary 18.6. So the Bethe equations are the condition that the ansatz produce an eigenvalue of the right kind: the apparent poles of (18.13) at the roots must cancel. For $M = 1$, $\Lambda Q = (\lambda + i)^N (\lambda - v - i) + \lambda^N (\lambda - v + i)$ is divisible by $\lambda - v$ iff $(v + i)^N = v^N$, that is $e^{ikN} = 1$ by (18.5): the magnon grid of (15.4).

Figure 18.4 shows both polynomials for two states met in Lecture 15. The zeros of Q are the roots; Λ , of degree N with leading coefficient 2, has its own zeros elsewhere. In Lecture 19 the roots of the ground state fill a line densely and the Bethe equations become an integral equation for their density, with Λ supplying the energies; Lecture 20 generalises the T-Q relation itself · *sketched there*.

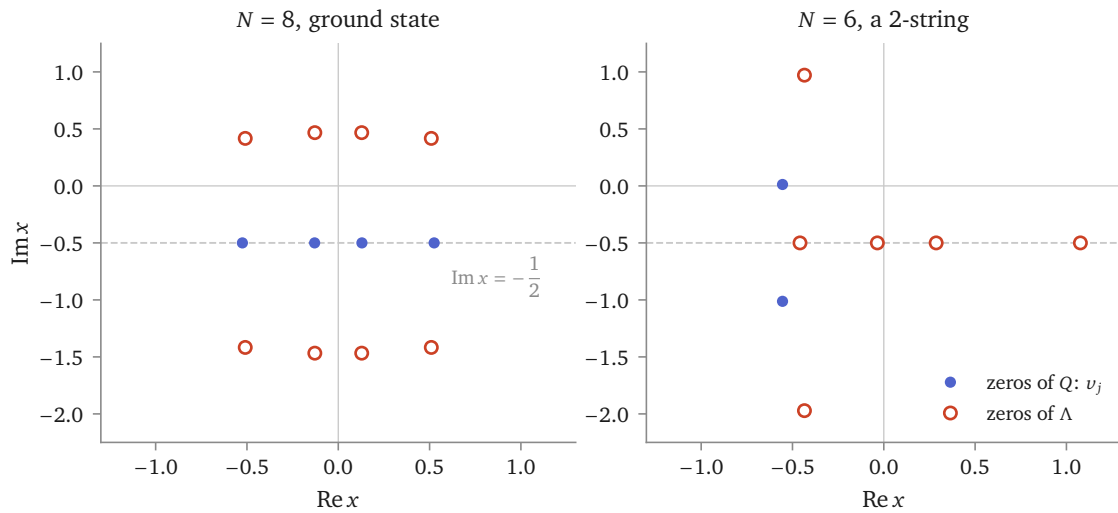


Figure 18.4: The two polynomials of the T-Q relation in the complex plane of their argument x . Left: the ground state of $N = 8$ ($M = 4$, real rapidities, so the roots lie on $\text{Im } x = -\frac{1}{2}$); the eight zeros of Λ lie in two rows, at distance about 1 above and below. Right: the 2-string of $N = 6$ at $K = 2\pi/3$ from Figure 15.4, whose roots $x \approx -\frac{1}{\sqrt{3}}$ and $-\frac{1}{\sqrt{3}} - i$ are nearly the singular pair shifted. In both, Λ is checked to be a polynomial and an eigenvalue of the exact transfer matrix at three complex points. (scripts/figures/tq_roots.py.)

In this form the equations can be asked of any polynomial Q , whether or not it comes from a nonzero Bethe vector. The completeness theorem cited in [Lecture 15](#) (Mukhin, Tarasov and Varchenko) is phrased in this language, for polynomial solutions of (18.18) with extra conditions that absorb the singular solutions.

18.6 The XXZ chain

The anisotropic chain of [Lecture 13](#),

$$H_{\text{XXZ}} = J \sum_{n=1}^N \left(S_n^x S_{n+1}^x + S_n^y S_{n+1}^y + \Delta (S_n^z S_{n+1}^z - \frac{1}{4}) \right), \quad J > 0, \quad (18.19)$$

is (15.1) at $\Delta = 1$. Every step of this lecture, and every step of [Lecture 17](#) from the *RLL* relation onwards, goes through with the linear functions replaced by hyperbolic ones (the *RLL* relation itself is a separate computation, since [Theorem 17.1](#) was proved inside the group algebra of the permutations of three objects) · *sketched: the derivation is §§ 18.1 to 18.4 line by line, with the changes listed in the reading at the end; each formula is checked by the check script.* In the basis $\uparrow\uparrow, \uparrow\downarrow, \downarrow\uparrow, \downarrow\downarrow$ the trigonometric *R*-matrix is

$$R(\lambda) = \begin{pmatrix} \sinh(\lambda - i\eta) & & & \\ & \sinh \lambda & -i \sin \eta & \\ & -i \sin \eta & \sinh \lambda & \\ & & & \sinh(\lambda - i\eta) \end{pmatrix}, \quad \Delta = \cos \eta. \quad (18.20)$$

It satisfies the Yang–Baxter equation (17.9), and as $\eta \rightarrow 0$ with $\lambda \rightarrow \eta\lambda$, $R/\eta \rightarrow \lambda - iP$. The Lax operator is again $L_{\text{an}}(\lambda) = -R_{\text{an}}(-\lambda)$, as in (17.8), and satisfies *RLL* in the course's ordering. On $|0\rangle$, $A = \sinh^N(\lambda + i\eta)$ and $D = \sinh^N \lambda$, and $B(\nu) |0\rangle$ is the magnon with $\nu = -\lambda - i\eta/2$, $e^{ik} = \sinh(\lambda + i\eta/2)/\sinh(\lambda - i\eta/2)$. The result:

$$\Lambda(\lambda) = \sinh^N(\lambda + i\eta) \prod_j \frac{\sinh(\lambda - \nu_j - i\eta)}{\sinh(\lambda - \nu_j)} + \sinh^N \lambda \prod_j \frac{\sinh(\lambda - \nu_j + i\eta)}{\sinh(\lambda - \nu_j)}, \quad (18.21)$$

and the Bethe equations and energy, in $\lambda_j = -\nu_j - i\eta/2$,

$$\left(\frac{\sinh(\lambda_j + i\eta/2)}{\sinh(\lambda_j - i\eta/2)} \right)^N = \prod_{k \neq j} \frac{\sinh(\lambda_j - \lambda_k + i\eta)}{\sinh(\lambda_j - \lambda_k - i\eta)}, \quad E = \sum_j \frac{J \sin^2 \eta}{\cos \eta - \cosh 2\lambda_j}. \quad (18.22)$$

At $\Delta = 0$ ($\eta = \pi/2$) the right side is $(-1)^{M-1}$: the chain is the XX chain of [Lecture 14](#), and the equations are the free-fermion grids (14.11), as that lecture said.

The formulas hold with η complex, and the regimes of [Figure 18.5](#) differ in where the roots of the ground state lie; for $\Delta < -1$ the reference state is a ground state (so is the all-down state). What each regime looks like in the thermodynamic limit is [Lecture 19](#)'s business for $\Delta = 1$ and is cited here for the rest · *cited: Franchini, §§4.3–4.5; Takahashi, ch. 4.*

18.7 The dictionary

In [Lecture 10](#) the eigenvalues z of the classical monodromy solve $z^2 - t(\lambda)z + \det T = 0$, i.e. $z + \det T/z = t(\lambda)$ with $\det T = 1$: the spectral curve ([Definition 10.8](#)); the eigenvectors are the Bloch solutions ([Proposition 10.7](#)). Write $z(\lambda) = \lambda^N Q(\lambda + i)/Q(\lambda)$; then (18.18) reads

$$\Lambda(\lambda) = z(\lambda) + \frac{\text{qdet } T(\lambda - i)}{z(\lambda - i)}, \quad \text{qdet } T(\mu) = [\mu(\mu + 2i)]^N, \quad (18.23)$$

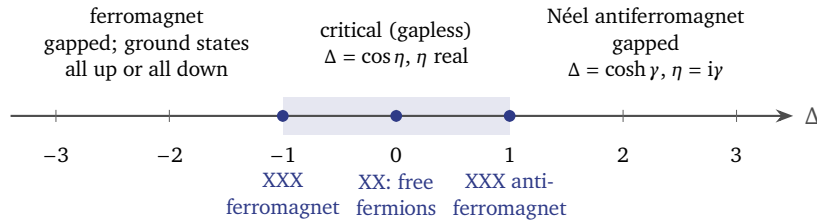


Figure 18.5: The regimes of the XXZ chain (18.19) with $J > 0$ at zero field. The spectrum is gapless for $-1 \leq \Delta \leq 1$ (the shaded interval is the critical regime) and has a gap above the ground states outside it. For $\Delta < -1$ the all-up state is a ground state (the check script, $N = 8$); the other labels are cited (Franchini, ch. 4, whose Δ is minus ours). At $\Delta = -1$ a rotation of every other spin (even N) turns the chain into the ferromagnetic Heisenberg chain.

with the quantum determinant of Lecture 17's reading in the place of $\det T$ and a shift by i where the classical curve has none (the check script, for general N and $M \leq 3$). The function Q plays the part of the eigenvector: a ratio of its values gives the multiplier z , as the ratio of a Bloch solution's values one period apart gave the Floquet multiplier. The analogy is made precise by Sklyanin's separation of variables, in which the eigenvector becomes a product of one-variable factors, each a Q · cited. It is not this chain's $B(\lambda)$ that separates the variables: by (18.2), B lowers S^z , so $B(\lambda)^{N+1} = 0$ and its spectrum is $\{0\}$. Inhomogeneities do not help — they leave (18.2) untouched — so the construction twists the monodromy by a numerical matrix and assumes its data generic; the homogeneous untwisted chain is a separate matter (Sklyanin 1995). That is one of the frontiers of Lecture 23.

Dictionary

The spectral curve $z + \det T/z = t(\lambda)$ of the periodic Toda chain (Definition 10.8) $\rightarrow$ Baxter's T–Q relation (18.18), as $\Lambda = z(\lambda) + q \det T(\lambda - i)/z(\lambda - i)$ (18.23) (type A; separation of variables makes the analogy precise for a twisted chain with generic data, cited)

The Floquet multiplier z and the Bloch eigenvector of $T(\lambda)$ (Proposition 10.7) $\rightarrow z(\lambda) = \lambda^N Q(\lambda + i)/Q(\lambda)$ and Baxter's Q , whose zeros are the Bethe roots (type A, C)

$\det T = 1$ for the Toda chain (10.13) $\rightarrow$ the quantum determinant $[\mu(\mu + 2i)]^N$, central, whose two factors are D 's vacuum eigenvalue at μ and A 's at $\mu + i$ (18.3) (type A)

Two more objects complete the method, and the reading below names them: the norm of a Bethe vector and its scalar product with a vector that is not an eigenvector. Correlation functions (optional lecture C) are built from them.

Reading

18.8 The XXZ chain, step by step

The changes to §§ 18.1 to 18.4 for the XXZ chain, each checked by the check script:

- *Regular point.* $t(0) = (i \sin \eta)^N U$ and $H_{\text{XXZ}} = \frac{J}{2} (\sin \eta \cdot i \frac{d}{d\lambda} \ln t|_0 - N \cos \eta)$.
- *Reference state.* $C|0\rangle = 0$ and $B(v)|0\rangle = i \sin \eta \sum_n \sinh^{n-1} v \sinh^{N-n}(v + i\eta)|n\rangle$, so for $\sinh v \neq 0 \neq \sinh(v + i\eta)$ the amplitudes are in the ratio $\sinh v / \sinh(v + i\eta) = e^{ik}$ from site to site (at the excluded values, as in § 18.1, the vector sits on one site); one magnon has $E = J(\cos k - \Delta)$.

- *Exchange relations.* (18.9) and (18.10) with $f(u) = \sinh(u - i\eta)/\sinh u$ and $g(u) = i \sin \eta/\sinh u$; the identity of Lemma 18.5's proof holds for these too.

The parametrisation depends on the regime. For $|\Delta| < 1$, η is real and the rapidities of the ground state are real; for $\Delta > 1$ one writes $\eta = i\gamma$, the hyperbolic functions of $\lambda \pm i\eta/2$ become trigonometric after $\lambda \rightarrow i\lambda$, and the rapidities live on a circle; for $\Delta < -1$ the reference state is a ground state. The formulas above hold in all three, with η complex (the check script samples $\Delta = \frac{1}{2}, -\frac{1}{2}, 2$).

Reading

18.9 Inhomogeneous chains

Nothing above used that every site has the same spectral parameter. With $L_{an}(\lambda - \xi_n)$ at site n , the RTT relation holds as before (Theorem 17.3 needs only a common R), the exchange relations are unchanged, and (18.3) becomes $\prod_n(\lambda - \xi_n + i)$ and $\prod_n(\lambda - \xi_n)$; so the Bethe equations and Λ follow with $(\nu + i)^N \rightarrow \prod_n(\nu - \xi_n + i)$ and $\nu^N \rightarrow \prod_n(\nu - \xi_n)$ (the check script, $N = 4$). This is the form Lecture 22 needs: there the sites are particles, $\xi_n = k_n/c$ are their momenta in units of the coupling, and the Lax operator is the fermions' two-body S -matrix $X_F = -L(u)/(u + i)$ of (16.14), up to the scalar.

Reading

18.10 Norms and scalar products

An eigenvector found by the Bethe ansatz comes unnormalised, and any expectation value needs its norm. For the wavefunction of Lecture 15 with *real* rapidities, normalised by $A_{\text{id}} = 1$ in (15.14), the answer is a determinant:

$$\sum_{x_1 < \dots < x_M} |\psi(x_1, \dots, x_M)|^2 = \det \left(\frac{\partial \Phi_j}{\partial k_l} \right)_{j,l=1}^M, \quad \Phi_j = Nk_j + \sum_{l \neq j} \theta_2(\lambda_j - \lambda_l), \quad (18.24)$$

the Jacobian of the Bethe equations in the form (15.20), with each λ_l a function of k_l · *cited: Gaudin 1983, proved by Korepin 1982; Korepin, Bogoliubov and Izergin, §§X.1–X.3; the check script verifies it for $N = 6, 7, 8$, $M \leq 4$. For $M = 1$ it says $\sum_x |e^{ikx}|^2 = N$. In the rapidities, with F_j the left side of the logarithmic equations (15.18), the same determinant is*

$$\det \left(\frac{\partial \Phi_j}{\partial k_l} \right) = \prod_l \left(\lambda_l^2 + \frac{1}{4} \right) \det \left(\frac{\partial F_j}{\partial \lambda_l} \right),$$

because $\Phi_j = N\pi - F_j$ and $dk/d\lambda = -\theta'_1(\lambda) = -4/(1 + 4\lambda^2)$, and the second matrix is symmetric — the Hessian of a function playing, on the lattice, the part of the convex function of Proposition 16.5. The function whose stationary points are the solutions also measures the size of every Bethe vector.

Real rapidities are a hypothesis, not a convenience. Normalising by $A_{\text{id}} = 1$ breaks the symmetry of ψ under relabelling the roots unless $|S(\lambda_j - \lambda_l)| = 1$, while the determinant is symmetric; so no such formula can hold for complex roots in this normalisation. The 2-string of Figure 18.4 is a counterexample: there $\sum |\psi|^2 = 2.16964\dots$ and the determinant is $176.2159\dots$, in the ratio $S(\lambda_1 - \lambda_2) = 0.0123\dots$ (the check script). For complex roots

the determinant needs the conjugation and ordering factors that the bare formula drops; summing $|\psi|^2$ directly of course still works, and the algebraic scalar product below is the systematic route.

Korepin's proof does that. The relevant object is the *algebraic scalar product*

$$\langle 0 | C(u_1) \cdots C(u_M) B(v_1) \cdots B(v_M) | 0 \rangle,$$

which the exchange relations for C with B (more entries of (18.6)) reduce to a sum over partitions of the roots (Korepin, Bogoliubov and Izergin, ch. IX). It is a bilinear pairing, not an inner product: in this course's conventions it equals $(-1)^M \|\Psi\|^2$ at coincident real rapidities (the check script), and the sources' relation $C(\bar{\lambda}) \propto B(\lambda)^\dagger$ belongs to their conventions, not to ours — Remark 17.2 once more. The proof itself compares two functions of M free parameters. Let the model be deformed so that the two vacuum eigenvalues $a(\lambda), d(\lambda)$ are arbitrary, and set

$$z_j = i \frac{d}{d\lambda} \ln \frac{a(\lambda)}{d(\lambda)} \Big|_{\lambda_j}, \quad K_{jl} = \theta'_2(\lambda_j - \lambda_l) = \frac{2}{1 + (\lambda_j - \lambda_l)^2},$$

so that for the chain $z_j = N/(\lambda_j^2 + \frac{1}{4})$ and the matrix above is $H_{jl} = \delta_{jl}(z_j - \sum_{m \neq j} K_{jm}) + (1 - \delta_{jl})K_{jl}$. Both the normalised scalar product — the pairing divided by $\prod_j a(\lambda_j)d(\lambda_j) \prod_{j \neq k} f(\lambda_j - \lambda_k)$ — and $\det H$ have five properties: they are invariant under exchanging the pairs (z_j, λ_j) ; linear in each z_j ; equal to 1 at $M = 0$ and to z_1 at $M = 1$; zero when every z_j is zero and $M \geq 1$; and their derivative in z_M is the same quantity for the remaining $M - 1$ roots *with the modified parameters* $\tilde{z}_j = z_j - K_{jM}$. The modification is the point: $\partial_{z_2} \det H = z_1 - K_{12}$, not z_1 . Two functions with all five agree, by induction on M : their difference has zero derivative in every z_j , hence is constant, and vanishes at $z = 0$ · *sketched; the properties and the induction are Korepin, Bogoliubov and Izergin, §§X.2–X.3, in their normalisation.*

When one of the two vectors is an eigenvector and the other is not, Slavnov (1989) found the scalar product itself to be a single determinant · *cited*. That formula is the gateway to form factors — matrix elements of S_n^z and $S_n^\pm$ between eigenstates — because the local spin operators can themselves be written through A, B, C, D (the quantum inverse problem, Kitanine, Maillet and Terras 1999). This is the subject of optional lecture C (Slavnov 2022 is the text).

18.11 Four sites by the algebraic Bethe ansatz

Take $N = 4$. *One magnon.* (18.16) with $M = 1$ is $((v + i)/v)^4 = 1$, so $v/(v + i) = e^{ik}$ with $k = \frac{\pi m}{2}$, $m = 1, 2, 3$ ($m = 0$ is $v = \infty$, the descendant $S^- |0\rangle$), i.e. $v = ie^{ik}/(1 - e^{ik})$: $v = -\frac{1}{2} - \frac{i}{2}$, $-\frac{i}{2}$, $\frac{1}{2} - \frac{i}{2}$. Proposition 18.7 gives $E = J(\cos k - 1) = -J, -2J, -J$, (15.4).

Two magnons. (15.2) asks for $\binom{4}{2} - \binom{4}{1} = 2$ highest-weight states. Try symmetric real rapidities $\lambda_{1,2} = \pm a$. (15.15) for $j = 1$ reads $(\frac{a+i/2}{a-i/2})^4 = \frac{2a+i}{2a-i}$, i.e. $(\frac{2a+i}{2a-i})^3 = 1$, whose real solution with $a > 0$ is $a = \frac{1}{2\sqrt{3}}$ (then $\frac{2a+i}{2a-i} = e^{2\pi i/3}$). So $v_{1,2} = \mp \frac{1}{2\sqrt{3}} - \frac{i}{2}$, and (18.13) simplifies to

$$\Lambda(\lambda) = 2\lambda^4 + 4i\lambda^3 + 2i\lambda + 1,$$

a polynomial, as Proposition 18.8 requires. Then $\Lambda(0) = 1 = i^4 e^{-iK}$ gives $K = 0$, and $\Lambda'(0)/\Lambda(0) = 2i$ gives $E = \frac{J}{2}(i \cdot 2i - 4) = -3J$: the ground state of the four-site ring, the singlet of total spin 0 (check: $\sum \mathbf{S}_n \cdot \mathbf{S}_{n+1} = -2$ there). The second highest-weight state is the

singular one, $E = -J$, $K = \pi$: its roots are the pair $\{0, -i\}$, for which $\Psi = 0$, and the algebraic Bethe ansatz does not produce it. Compare Table 15.1, where $N = 6$ has the same structure: every regular solution, plus one singular.

Summary.

- *The reference state:* $A|0\rangle = (\lambda + i)^N |0\rangle$, $D|0\rangle = \lambda^N |0\rangle$, $C|0\rangle = 0$, and B creates: for $v \neq 0, -i$, $B(v)|0\rangle$ is the magnon with $e^{ik} = v/(v + i)$, so $v = -\lambda - \frac{i}{2}$ in the rapidity of Lecture 15 (Proposition 18.1).
- *Exchange relations,* three entries of RTT : $[B, B] = 0$, and $A(\lambda)B(\mu)$, $D(\lambda)B(\mu)$ as (18.9), (18.10) (Proposition 18.3).
- *Wanted and unwanted terms:* $t(\lambda)$ on $\Psi = \prod B(v_j)|0\rangle$ gives $\Lambda\Psi$ plus terms with one v_j replaced by λ (Theorem 18.4); these cancel iff the Bethe equations (18.16) hold, which are (15.15) (Corollary 18.6). Singular roots can give $\Psi = 0$; completeness is cited.
- *The eigenvalue:* $\Lambda(\lambda)$ (18.13) gives the momentum and the energy of Lecture 15 (Proposition 18.7).
- *T-Q:* $\Lambda Q = (\lambda + i)^N Q(\lambda - i) + \lambda^N Q(\lambda + i)$; the Bethe equations say Λ is a polynomial (Proposition 18.8); it is the quantum form of the spectral curve.
- *XXZ:* the same steps with $\sinh$; $\Delta = \cos \eta$ (§ 18.6).

Tutorial. Exercise Sheet 18.

References

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